

## Peer Review

# Review of: "Accuracy Without Profit: A Statistical Evaluation of Machine Learning Profitability in the English Premier League"

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The paper looks at whether the use of well-known machine learning algorithms can help find profitable forecasts in the English Premier League betting market. In general, the paper finds that statistical accuracy is reasonable, but this accuracy does not translate into profits against a bookmaker (reliably) over a long sample period. While it is interesting to see the relative performance of the models considered here (some well-known and implementable models such as *xGBoost* and *randomforest*, the results are not overly surprising.

Any gambling strategy is going to have some variance season to season, as a consequence of uncertain outcomes – and a well-known deficiency of the Kelly criterion is that it might overbet in situations of greater certainty of outcomes. The most interesting plot in the paper is Figure 4(b), which shows that the models perform quite poorly in terms of calibration, particularly when odds are high. In other words, the models tend to perform poorly on extreme favourites, which is precisely when the Kelly criterion will assign a large stake. This is one of the main motivations for ‘fractional-Kelly’ strategies, or more precise bankroll management.

Some other aspects are worth considering for future research. Firstly, modern predictions would use more complex variables (as analytics has improved substantially since 2001). It would be very useful to know whether they can help to beat the bookmaker (at least in some subsample). It may be that other data could provide useful insights into whether a draw is likely – consider things like Expected Goals or player-specific variables. Secondly, correct application of the Kelly criterion will require consideration of ‘diversifying bets’ on the draw outcome (in addition to home/away). I would have liked to see this in the

breakdown of profitability. Similarly, issues such as games kicking off simultaneously (which will affect staking amounts), cross-market arbitrage, bookmaker pricing strategies, and model retraining would be helpful to discuss. There is a lot of literature on sports betting that covers these topics.

Overall, the paper does highlight some key aspects of simply applying machine learning and expecting that statistical accuracy translates into profitability. It is not just accuracy in prediction, but how the decision-maker employs the probabilities that makes a difference.

## **Declarations**

**Potential competing interests:** No potential competing interests to declare.